

HOI Query
Person: **holding**
Object: bottle

HOI Query
Person: **riding**
Object: horse

...
HOI Query
× 600

LLM Optimization Agent

You are an expert VDMS Optimization Engine for HOI Retrieval.
Iteration: 4/30 | Phase: EXPLORATION | Best Score: 3.79 | Score = mAP × QPS

▼ **Failures:** [M=16, efS=32 → 2.14], [k=500, α=0.30 → 1.92] ★ **Sweet Spots:** [M=32, efS=64, k=200, α=0.50 → Score=3.79]



Iteration 1: FaissHNSWFlat | M=32, efSearch=100, k=200, α=0.50, n_refs=3, centroid
Score: 3.75 | mAP: 0.2233 | QPS: 16.8
Reasoning: efSearch=100 + k=200 balances speed with recall; M=32 ensures HNSW graph connectivity.



Iteration 2: FaissHNSWFlat | M=16, efSearch=32, k=500, α=0.30, n_refs=1, first
Score: 2.14 | mAP: 0.1124 | QPS: 19.1 ✖ **LOW SCORE** — logged to failure history
M=16 reduces graph connectivity; efSearch=32 insufficient for recall. Marked as failure pattern.



Iteration 3: FaissHNSWFlat | M=32, efSearch=64, k=200, α=0.50, n_refs=3, centroid
Score: 3.79 ★ (Best so far) | mAP: 0.2233 | QPS: 17.0
Reasoning: efSearch: 100 → 64 boosts QPS while preserving recall; M=32 and k=200 maintained.

Iterations ...

► **Query to LLM:** Given iteration history, failure patterns, and current best (Score=3.79), suggest the next configuration to maximize Score = mAP × QPS.

LLM Response (Iteration 4):

```
{ "engine": "FaissHNSWFlat", "params": { "M": 32, "efSearch": 100, "efConstruction": 200 },  
  "k_neighbors": 200, "alpha": 0.50, "n_refs": 3, "ref_strategy": "centroid",  
  "reasoning": "Return to efSearch=100 to recover mAP; lower efSearch reduces recall." }
```